

DEVELOPMENT OF AN INTERACTIVE WEB PLATFORM FOR NUMERICAL ANALYSIS



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Abstract

The aim of this project is to develop "ALGO," a web application designed to meet the needs of students worldwide by providing a centralized platform for learning and practicing numerical analysis. ALGO aims to change students' perceptions of numerical analysis as a difficult subject by offering a user-friendly interface that enhances their academic performance. Through intuitive interfaces and dependable backend systems, the application offers a reliable and convenient solution for mastering numerical analysis. This documentation outlines the objectives, design, development, and testing processes involved in creating ALGO.

Introduction

Our thesis focuses on developing a web application designed to facilitate the easy solving of numerical analysis problems. This web application, named "ALGO," is derived from the word "algorithm," which refers to a step-by-step procedure for solving a problem or achieving a specific goal. ALGO is primarily designed for students and professionals, including engineers, scientific computing experts, data analysts, statisticians, economists, and those involved in financial modeling.

Problem Statement

Numerical analysis is a fundamental aspect of many academic disciplines, yet it is often perceived as challenging and inaccessible by students worldwide. This negative perception can hinder students' academic performance and their willingness to engage with the subject. There is a pressing need for an intuitive, comprehensive web application that can provide students with the resources and support they need to overcome these challenges and succeed in their studies.

Project Aim

The aim of this project is to develop "ALGO," a comprehensive web application designed to support students worldwide in mastering numerical analysis. By providing a centralized platform with user-friendly interfaces and dependable backend systems, ALGO seeks to transform students' perceptions of numerical analysis from a challenging subject to an accessible and manageable one.

Objective

- Develop a User-Friendly Interface
- Provide Comprehensive Learning Resources
- Implement Reliable Backend Systems
- Enhance Academic Performance
- Expand Accessibility

Scope of the project

- **Target Audience:** Students and professionals in engineering, scientific computing, data analysis, statistics, economics, and financial modeling.
- **Functionality:** Features like tutorials, interactive exercises, problem-solving tools, and performance tracking.
- **Geographical Reach:** Accessible worldwide with multilingual support.
- **Usability Testing:** Rigorous testing to ensure user-friendliness, reliability, and effectiveness.

Approach

1. **Research and Planning:** Thorough research on the needs of students and professionals in numerical analysis.
2. **Design and Development:** Create wireframes and prototypes, followed by development using appropriate technologies.
3. **Content Creation:** Develop comprehensive learning materials.
4. **Testing and Feedback:** Conduct usability testing to gather feedback and make improvements.
5. **Deployment and Maintenance:** Deploy the application and ensure ongoing maintenance and updates.

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